TRIPOD-AI, fairness auditing, reproducibility at scale

Workflow labs use the variant template: Goal → Approach → Execution → Check → Report.

Learning objectives

- Enumerate the TRIPOD-AI reporting items relevant to a prediction- model manuscript.
- Compute group-stratified AUC and calibration as a fairness audit.
- Sketch a reproducible analysis pipeline with the `targets` package.

Prerequisites

External validation; biomarker evaluation.

Background

TRIPOD-AI extends the original TRIPOD statement to cover machine- learning prediction models. It asks authors to describe the data source, the participants, the outcome, the predictors, sample size and missing data, the model specification and its hyperparameter tuning, the performance on internal and external data, and the intended use of the model. A report that fails on any of these items is difficult to reproduce and difficult to deploy safely.

Fairness auditing extends validation to population subgroups. A model with strong overall AUC can have markedly worse performance in a minority subgroup; the remedy is first to detect the gap and then to decide whether to retrain, reweight, or accept the limitation explicitly.

The `targets` package is the modern R approach to reproducible pipelines. It builds a directed acyclic graph of analysis steps, caches intermediate outputs, and reruns only what has changed. This separation between pipeline definition and execution is what lets a study survive the months between submission and revision.

Reproducibility at scale is not a purity test. It is an insurance policy: when a reviewer asks for a recomputed sensitivity, or when a colleague tries to replicate the analysis two years later, the cost of doing the work as a scripted DAG is paid back many times.

Setup

```r
library(tidyverse)
library(pROC)
library(MASS)
set.seed(42)
theme_set(theme_minimal(base_size = 12))
```

1. Goal

Audit a logistic prediction model on `Pima.tr` by a simulated subgroup attribute, and sketch a `targets` pipeline for the full analysis.

2. Approach

Attach a synthetic subgroup label — imagine this were clinic of enrolment — and compare performance.

```r
  mutate(subgroup = sample(c("A", "B"), n(), replace = TRUE,
                           prob = c(0.7, 0.3)))
ggplot(d, aes(glu, fill = subgroup)) +
  geom_histogram(alpha = 0.7, bins = 20, position = "identity")
```

3. Execution

```r
d$p <- predict(fit, type = "response")

auc_overall <- as.numeric(auc(roc(d$type, d$p, quiet = TRUE)))
auc_by <- d |>
  group_by(subgroup) |>
  summarise(auc = as.numeric(auc(roc(type, p, quiet = TRUE))),
            n   = n(), .groups = "drop")
auc_by
```

Calibration stratified by subgroup.

```r
  mutate(bin = cut(p, quantile(p, seq(0, 1, by = 0.2)),
                   include.lowest = TRUE)) |>
  group_by(subgroup, bin) |>
  summarise(pred = mean(p), obs = mean(type == "Yes"),
            n = n(), .groups = "drop") |>
  ggplot(aes(pred, obs, colour = subgroup)) +
  geom_point(aes(size = n)) + geom_line() +
  geom_abline(slope = 1, intercept = 0, colour = "grey50") +
  labs(x = "mean predicted", y = "observed proportion")
```

A minimal `targets` pipeline (sketch).

```r
library(targets)
tar_script({
  library(tidyverse); library(MASS); library(pROC)
  list(
    tar_target(raw, as_tibble(MASS::Pima.tr)),
    tar_target(fit, glm(type ~ glu + bmi + age, data = raw, family = binomial())),
    tar_target(auc_overall,
               as.numeric(auc(roc(raw$type, predict(fit, type = "response"), quiet = TRUE)))),
    tar_target(report, tibble(auc = auc_overall))
  )
})
tar_make()
tar_read(report)
```

4. Check

TRIPOD-AI-style checklist (abbreviated).

```r
  ~item,                              ~status,
  "Study design stated",              "yes",
  "Source and eligibility",           "yes",
  "Outcome definition",               "yes",
  "Predictor definitions",            "yes",
  "Sample size justified",            "partial",
  "Missing-data handling",            "yes",
```

```
  "Model specification",             "yes",
  "Hyperparameter tuning",           "NA (no tuning)",
  "Internal validation",             "yes",
  "External validation",             "NOT in this lab",
  "Calibration reported",            "yes",
  "Fairness audit by subgroup",      "yes",
  "Code available",                  "yes"
)
checklist
```

5. Report

> A logistic prediction model on `Pima.tr` achieved overall AUC `round(auc_overall, 2)`. A fairness audit by synthetic subgroup revealed AUCs of `round(auc_by$auc[1], 2)` in subgroup A (n = `auc_by$n[1]`) and `round(auc_by$auc[2], 2)` in subgroup B (n = `auc_by$n[2]`). A `targets` pipeline capturing raw data, fit, evaluation, and report would make the entire analysis re-runnable by any collaborator.

TRIPOD-AI, fairness auditing, and a pipeline tool are not independent initiatives; they are three faces of the same commitment to make modelling decisions legible, auditable, and reproducible.

Common pitfalls

- Reporting overall metrics and stopping; fairness gaps are only visible after stratification.
- Using `targets` as a static pipeline and not updating the DAG when inputs change.
- Treating TRIPOD-AI as a post-hoc checklist rather than a planning document written before analysis.

Further reading

- Collins GS et al. (2024), *TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods.*
- Obermeyer Z et al. (2019), *Dissecting racial bias in an algorithm used to manage the health of populations.*
- Landau WM (2021), *The targets R package: a dynamic make-like function-oriented pipeline toolkit.*

Session info

```

```

See also — chapter index

- Methods in this chapter
- Find your method (Chapter 0)